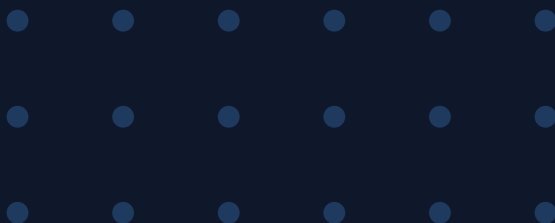


eScience Institute, University of Washington

LLMoxie Analytics Pipeline

Understanding how LLMs are used
in research software engineering

Bhagyashree Wagh · May 2026



We're logging every LLM call.

But what are we actually learning from it?



Raw data, no structure

Thousands of individual JSON files with no sessions, no easy querying, no aggregation.



Privacy risk

Real user identities appear in raw logs. No anonymization before storage or analysis.



No visibility

No way to see which models are used most, who's spending what, or when usage spikes.

What this pipeline unlocks

From raw logs to actionable research intelligence



Proper Sessions

Group individual requests into sessions to see full conversation context, not isolated calls.



Anonymized Data

Real identities replaced with consistent anonymous IDs. Reversible internally by admins.



Dashboard + Querying

Live Streamlit dashboard with cost, usage, latency, and session analytics across 6 views.



Fine-tuning Datasets

Export curated interaction data to generate datasets for model fine-tuning.

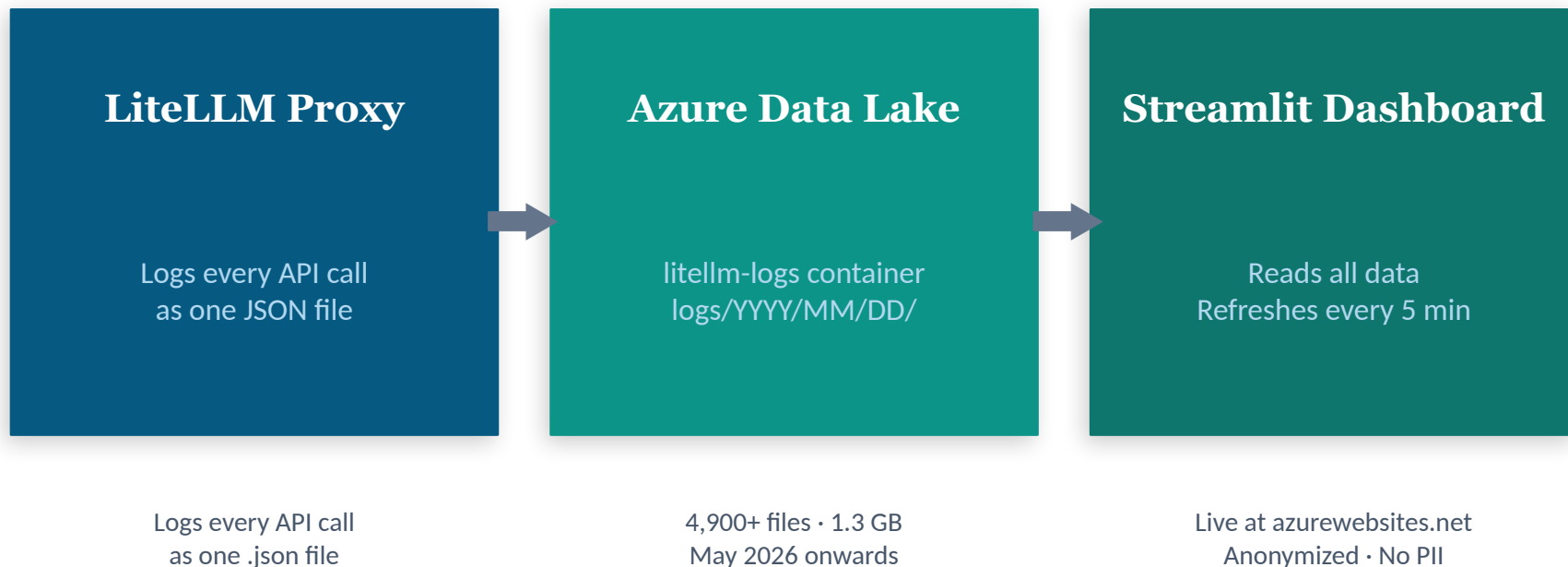


Evaluate Tool Use

Understand how researchers use tools, skills, and MCP servers in practice.

How it works

One pipeline, three steps



What's been built

- ✓ Live dashboard connected to Azure Data Lake
- ✓ Reads all historical data (4,900+ log files)
- ✓ 6 analytics tabs: Overview, Cost, Time, Users, Sessions, AI Q&A
- ✓ Reversible PII anonymization (user_001, user_002...)
- ✓ Reverse mapping stored securely in Azure Key Vault
- ✓ Auto-refresh every 5 minutes, always current
- ✓ Deployed on Azure App Service with managed identity

STATUS

Completed

4,900+
log files

6
dashboard tabs

0 PII
exposed

5 min
refresh cycle

The dashboard: 6 views

1

Overview

Total spend, requests, tokens and error rate at a glance

2

Cost Explorer

Spend by model, team and user to break down the bill

3

Time Intelligence

Hourly and daily patterns showing when researchers are most active

4

Users

Per-user spend and activity (anonymized: user_001, user_002, ...)

5

Sessions

Multi-turn conversations grouped by session ID

6

Ask the Data

Natural language Q&A powered by Claude



DIRECTION

What's next



Historical data ingestion

Full historical data (8-day dump) being loaded; dashboard will show the complete picture



Fine-tuning dataset generation

Export anonymized session data in training format for model fine-tuning



Tool and MCP usage evaluation

Analyze which tools, skills, and MCP servers researchers rely on most



Programmatic access

Query API so teams can pull usage data into notebooks and reports